EQUAZIONI IN $\mathbb{C}$

$$z^2 - 2iz - 1 + 9i = 0$$

$$z_{1,2} = i \pm \sqrt{-1 + 1 - 9i} = \frac{2i \pm \sqrt{-4+4-36i}}{2}$$

$$= \frac{2i \pm 6\sqrt{-i}}{2} = i \pm 3\sqrt{-i}$$

Calcolo $\sqrt{-i}$

$$-i = \cos\frac{3}{2}\pi + i\sin\frac{3}{2}\pi$$

$$\sqrt{-i} = \left(\cos\frac{\frac{3}{2}\pi + 2k\pi}{2} + i\sin\frac{\frac{3}{2}\pi + 2k\pi}{2}\right)$$

$k = 0, 1$

$k=0$ $\quad \omega_0 = \cos\frac{3}{4}\pi + i\sin\frac{3}{4}\pi =$

$$= -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

$k=1$ $\quad \omega_1 = \cos\frac{7}{4}\pi + i\sin\frac{7}{4}\pi$

$$= \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$$

1) $z = i + 3\sqrt{-i} = \begin{cases} i + 3\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ i + 3\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \end{cases}$

2) $z = i - 3\sqrt{-i} = \begin{cases} i - 3\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ i - 3\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \end{cases}$

1) $z_1 = i - 3\frac{\sqrt{2}}{2} + 3i\frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2} + i\left(1 + \frac{3\sqrt{2}}{2}\right)$

$z_2 = i + 3\frac{\sqrt{2}}{2} - 3i\frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2} + i\left(1 - \frac{3\sqrt{2}}{2}\right)$

2) $z_3 = i + 3\frac{\sqrt{2}}{2} - 3i\frac{\sqrt{2}}{2} = z_2$

$z_4 = i - 3\frac{\sqrt{2}}{2} + 3i\frac{\sqrt{2}}{2} = z_1$

$\left(\frac{z+1}{z-2}\right)^3 = -8 \qquad w = \frac{z+1}{z-2}$

$w^3 = -8 \qquad w = \sqrt[3]{-8}$

$-8 = 8(\cos\pi + i\sin\pi)$

$w_k = \sqrt[3]{-8} = \sqrt[3]{8}\left(\cos\frac{\pi+2k\pi}{3} + i\sin\frac{\pi+2k\pi}{3}\right)$

$k = 0, 1, 2$

$k = 0 \quad w_0 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) =$

$= 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$

$k = 1 \quad w_1 = 2(\cos\pi + i\sin\pi) = -2$

$k = 2 \quad w_2 = 2\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) =$

$2\left(\frac{1}{2} - i\sqrt{3}/2\right)$

1) $w_0 = 1 + i\sqrt{3}$ $\qquad w = \frac{z+1}{z-2}$ $\qquad z \neq 2$

$1 + i\sqrt{3} = \frac{z+1}{z-2}$

$z + iz\sqrt{3} - 2 - 2i\sqrt{3} - z - 1 = 0$

$iz\sqrt{3} = 3 + 2i\sqrt{3}$

$z = \frac{3 + 2i\sqrt{3}}{i\sqrt{3}} \quad \frac{i\sqrt{3}}{i\sqrt{3}} = \frac{3i\sqrt{3} - 6}{-3}$

$$\boxed{z = 2 - i\sqrt{3}}$$

2) $w_1 = -2 \qquad \frac{z+1}{z-2} = -2$

$z + 1 + 2z - 4 = 0 \qquad 3z = 3$

$$\boxed{z = 1}$$

3) $w_2 = 1 - i\sqrt{3}$

$\frac{z+1}{z-2} = 1 - i\sqrt{3}$

$z + 1 - z + iz\sqrt{3} + 2 - 2i\sqrt{3} = 0$

$iz\sqrt{3} = -3 + 2i\sqrt{3}$

$z = \frac{-3 + 2i\sqrt{3}}{i\sqrt{3}} \cdot \frac{i\sqrt{3}}{i\sqrt{3}}$

$= \frac{-3i\sqrt{3} - 6}{-3} =$ $\boxed{2 + i\sqrt{3}}$

$z|z| = 2\bar{z}$

$z = \rho e^{i\theta}$

$|z| = \rho$

$\bar{z} = \rho e^{-i\theta}$

$(\rho e^{i\theta}) \cdot \rho = 2\rho e^{-i\theta}$

$\rho^2 e^{i\theta} = 2\rho e^{-i\theta}$

$\begin{cases} \rho^2 = 2\rho \\ \theta = -\theta + 2k\pi \end{cases}$ $k = 0, 1$

$\begin{cases} \rho^2 = 2\rho \\ 2\theta = 2k\pi \Rightarrow \theta = k\pi \end{cases}$

$\rho(\rho - 2) = 0 \begin{cases} \rho = 0 \\ \rho = 2 \end{cases}$

$\Rightarrow$ $z = 0$ soluzione

se $\rho = 2$ $\begin{cases} \theta = 0 \\ \theta = \pi \end{cases}$ $z = 2e^{i0} = 2$

$z = 2e^{i\pi}$

$iz = 3|z|^2 z$

$z = \rho e^{i\theta}$

$i\rho e^{i\theta} = 3\rho^2 \cdot \rho e^{-i\theta}$

$i = e^{i\pi/2}$

$\rho e^{i\pi/2} \cdot e^{i\theta} = 3\rho^3 e^{-i\theta}$

$\rho e^{i(\pi/2 + \theta)} = 3\rho^3 e^{-i\theta}$

$$\begin{cases} \rho = 3\rho^3 \\ \frac{\pi}{2} + \theta = -\theta + 2k\pi \end{cases} \qquad \begin{cases} \rho = 3\rho^3 \\ 2\theta = -\frac{\pi}{2} + 2k\pi \end{cases}$$

$$\begin{cases} \rho = 3\rho^3 \\ \theta = -\frac{\pi}{4} + k\pi \end{cases} \qquad k = 0, 1$$

1) $\rho = 0 \Rightarrow z = 0$

2) $\begin{cases} \rho^2 = \frac{1}{3} \Rightarrow \rho = \sqrt{\frac{1}{3}} = \frac{\sqrt{3}}{3} \\ \theta = -\frac{\pi}{4} + k\pi \quad k = 0, 1 \end{cases}$

$$z = \frac{\sqrt{3}}{3} e^{-i\pi/4} \; ; \quad z = \frac{\sqrt{3}}{3} e^{i\,3/4\,\pi}$$

$$z = \frac{\sqrt{3}}{3}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\pi/4\right)\right) =$$

$$= \frac{\sqrt{3}}{3}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{6}}{6} - i\frac{\sqrt{6}}{6}$$

$$z = \frac{\sqrt{3}}{3}\left(\cos\frac{3}{4}\pi + i\sin\left(\frac{3}{4}\pi\right)\right)$$

$$\frac{\sqrt{3}}{3}\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{6}}{6} + i\frac{\sqrt{6}}{6}$$

1) $4z = i\,|z|^2\,\bar{z}$ $\quad (z=0,\ z_0=\sqrt{2}+i\sqrt{2},\ z_1=-\sqrt{2}-i\sqrt{2})$

2) $(z-1)^3=-8i$ $\quad (z_0=1+2i,\ z_1=(-\sqrt{3}+1)-i,\ z_2=(\sqrt{3}+1)-i)$

Limiti di successioni

Verificare con la definizione i seguenti limiti:

1) $\lim_{n\to+\infty} \frac{n}{n^2+2}=0$ $\qquad a_n=\frac{n}{n^2+2}$ $\quad l=0$

2) $\lim_{n\to+\infty} \log n = +\infty$

3) $\lim_{n\to+\infty} \left(\frac{1}{3}\right)^n = 0$

1) $\forall \varepsilon>0 \ \exists N=N(\varepsilon):\ n>N \Rightarrow$

$$\left|\frac{n}{n^2+2}\right|<\varepsilon$$

$$|a_n-l|<\varepsilon$$

$$\frac{n}{n^2+2}<\varepsilon \qquad \frac{n-n^2\varepsilon-2\varepsilon}{n^2+2}<0$$

$n^2+2>0 \quad \forall n \in \mathbb{N}$

$n - n^2\varepsilon - 2\varepsilon < 0$

$n^2\varepsilon - n + 2\varepsilon > 0$

$n = \frac{1 \pm \sqrt{1-8\varepsilon^2}}{2\varepsilon}$

$\left(n < \frac{1-\sqrt{1-8\varepsilon^2}}{2\varepsilon} \right) \vee n > \underbrace{\frac{1+\sqrt{1-8\varepsilon^2}}{2\varepsilon}}_{N(\varepsilon)}$

2) $\lim_{n \to +\infty} \log n = +\infty$

$\forall M > 0 \quad \exists N = N(M) : n > N$

$\Rightarrow \log n > M$

$n > \underbrace{e^M}_{N(M)}$

3) $\lim_{n \to +\infty} \left(\frac{1}{3}\right)^n = 0$

$\forall \varepsilon > 0 \quad \exists N = N(\varepsilon) : n > N$

$\Rightarrow \left| \left(\frac{1}{3}\right)^n \right| < \varepsilon$

$\left(\frac{1}{3}\right)^n < \varepsilon \qquad n > \underbrace{\log_{\frac{1}{3}} \varepsilon}_{N(\varepsilon)}$

CALCOLO DI LIMITI DI SUCCESSIONI

1) $\lim_{n\to+\infty} \sqrt[n]{2^n+3^n} = \lim_{n\to+\infty} \sqrt[n]{3^n\cdot\left(\frac{2^n}{3^n}+1\right)}$

$= \lim_{n\to+\infty} 3\sqrt[n]{\left(\frac{2}{3}\right)^n+1} = 3$



2) $\lim_{n\to+\infty} \frac{\sqrt{n}-n+n^2}{2n^2-n^{3/2}+1} =$

$\lim_{n\to+\infty} \frac{n^2}{2n^2} = \frac{1}{2}$

3) $\lim_{n\to+\infty} \frac{\sqrt{n-\sin(n)}}{n+\cos n^2} = \lim_{n\to+\infty} \frac{\sqrt{n}}{n} = 0$

4) $\lim_{n\to+\infty} \left(1+\frac{1}{3n}\right)^{2n}$

$\lim_{n\to+\infty} \left(1+\frac{1}{n}\right)^{n} = e$

$\lim_{n\to+\infty} \left(1+\varepsilon_n\right)^{\frac{1}{\varepsilon_n}} = e \qquad \varepsilon_n \to 0$

$\lim_{n\to+\infty} \left[\underbrace{\left(1+\frac{1}{3n}\right)^{3n}}_{e}\right]^{\frac{2}{3}} = e^{2/3}$

b) $\lim_{n\to+\infty} \left(\frac{n^2+n}{n^2-n+2}\right)^n$

$\lim_{n\to+\infty} \left(\frac{(n^2-n+2)+2n-2}{n^2-n+2}\right)^n$

$\lim_{n\to+\infty} \left(1+\frac{2n-2}{n^2-n+2}\right)^n$ ($\frac{2n-2}{n^2-n+2} \to 0$)

$\lim_{n\to+\infty} \left[\left(1+\frac{2n-2}{n^2-n+2}\right)^{\frac{n^2-n+2}{2n-2}}\right]^{\frac{2n-2}{n^2-n+2}\cdot n}$

$\frac{2n^2-2n}{n^2-n+2}$

$\lim_{n\to+\infty} e^{\frac{2n^2-2n}{n^2-n+2}} = e^2$

c) $\lim_{n\to+\infty} e^n - n^{\sqrt{n}} = \lim_{n\to+\infty} e^n\left(1-\frac{n^{\sqrt{n}}}{e^n}\right)$

$n^{\sqrt{n}} = e^{\log n^{\sqrt{n}}}$ $\quad a = b^{\log_b a}$

$\lim_{n\to+\infty} e^n\left(1-\frac{e^{\log n^{\sqrt{n}}}}{e^n}\right)$

$\lim_{n\to+\infty} e^n\left(1-\frac{e^{\sqrt{n}\log n}}{e^n}\right)$

$$\lim_{n\to+\infty} e^n \left(1 - e^{\sqrt{n}\log n - n}\right)$$

$$\lim_{n\to+\infty} \sqrt{n}\log n - n =$$

$$\lim_{n\to+\infty} n\left(\underbrace{\frac{\log n}{\sqrt{n}}}_{\to 0} - 1\right) = -\infty$$

$$\lim_{n\to+\infty} e^n \left(1 - \underbrace{e^{\sqrt{n}\log n - n}}_{\to 0}\right) = +\infty$$

7) $$\lim_{n\to+\infty} \log(n^2+\sqrt{n}) - \log n$$

$$\lim_{n\to+\infty} \log\left(\frac{n^2+\sqrt{n}}{n}\right) = \lim_{n\to+\infty} \log \frac{n^2}{n}$$

$$= +\infty$$

8) $$\lim_{n\to+\infty} n \log\left(\frac{n+1}{n^2+2}\right) =$$

$$\lim_{n\to+\infty} n\left[\log(n+1) - \log(n^2+2)\right]$$

$$\lim_{n\to+\infty} n\left[\log n - \log n^2\right]$$

$$\lim_{n\to+\infty} n\left[\log n - 2\log n\right]$$

$$\lim_{n\to+\infty} -n\log n = -\infty$$

g) $$\lim_{n\to+\infty} \sqrt{4n^2-3n} - 2n + 1$$

$$\lim_{n\to+\infty} \sqrt{4n^2} - 2n = \lim_{n\to+\infty} 2n - 2n = 0$$

SBAGLIATO

$$\lim_{n\to+\infty} \frac{\sqrt{4n^2-3n} - (2n-1)}{\sqrt{4n^2-3n} + (2n-1)} \cdot \frac{\sqrt{4n^2-3n} + (2n-1)}{\sqrt{4n^2-3n} + (2n-1)}$$

$$\lim_{n\to+\infty} \frac{4n^2 - 3n - (4n^2 - 4n + 1)}{\sqrt{4n^2-3n} + (2n-1)}$$

$$\lim_{n\to+\infty} \frac{n-1}{2n+2n} = \lim_{n\to+\infty} \frac{n}{4n} = \frac{1}{4}$$

$$\lim_{n\to+\infty} \frac{n!}{n^n} = 0$$

CRITERIO DEL RAPPORTO

$\{a_n\}$ $a_n > 0$ $\lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = l$

se $l < 1 \Rightarrow a_n \to 0$; $l = 1$ caso dubbio

se $l > 1$ $a_n \to +\infty$

$$\lim_{n\to+\infty} \frac{n!}{n^n} = 0$$

$$\lim_{n\to+\infty} \frac{\dfrac{(n+1)!}{(n+1)^{(n+1)}}}{\dfrac{n!}{n^n}} =$$

$$\lim_{n\to+\infty} \frac{(n+1)\cdot n!}{(n+1)^n \cdot (n+1)} \cdot \frac{n^n}{n!}$$

$$\lim_{n\to+\infty} \left(\frac{n}{n+1}\right)^n = \lim_{n\to+\infty} \left(\frac{n+1}{n}\right)^{-n}$$

$$\lim_{n\to+\infty} \left[1 + \frac{1}{n}\right]^{-n} = \lim_{n\to+\infty} \left[\left(1+\frac{1}{n}\right)^n\right]^{-1}$$

$$= e^{-1} < 1 \Rightarrow \lim_{n\to+\infty} \frac{n!}{n^n} = 0$$

$$\lim_{n\to+\infty} \frac{(2n)!}{4^n}$$

$$\lim_{n\to+\infty} \frac{(2(n+1))!}{4^{n+1}} \cdot \frac{4^n}{(2n)!}$$

$$\lim_{n\to+\infty} \frac{(2n+2)!}{4^n \cdot 4} \cdot \frac{4^n}{(2n)!}$$

$$\lim_{n\to+\infty} \frac{(2n+2)(2n+1)\cancel{(2n)!}}{4} \cdot \frac{1}{\cancel{(2n)!}}$$

$$= +\infty \Rightarrow a_n \to +\infty$$

$$\lim_{n\to+\infty} \frac{n! + 2^n}{(n+2)!}$$

$$\lim_{n\to+\infty} \frac{(n+1)! + 2^{(n+1)}}{(n+3)!} \quad \frac{(n+2)!}{n! + 2^n}$$

$$\lim_{n\to+\infty} \frac{(n+1)n! + 2 \cdot 2^n}{(n+3)\cancel{(n+2)!}} \cdot \frac{\cancel{(n+2)!}}{n! + 2^n}$$

$$\lim_{n\to+\infty} \frac{n!\left((n+1)+2\cdot\frac{2^n}{n!}\right)}{(n+2)!\left((n+3)+(n+3)\frac{2^n}{n!}\right)}$$

$$\lim_{n\to+\infty} \frac{(n+1)+2\cdot\frac{2^n}{n!}}{(n+3)\left(1+\frac{2^n}{n!}\right)}$$

$$\lim_{n\to+\infty} \frac{(n+1)\left(1+\frac{2\cdot 2^n}{(n+1)n!}\right)}{(n+3)\left(1+\frac{2^n}{n!}\right)} = 1$$

($\frac{2\cdot 2^n}{(n+1)n!} \to 0$, $\frac{2^n}{n!} \to 0$)

CASO DUBBIO

$$\lim_{n\to+\infty} \frac{n!\left(1+\frac{2^n}{n!}\right)}{(n+2)!}$$